

Alec Sanchez — Security Researcher

Detection Engineering & Formal Assurance
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Independent security researcher working at the boundary between detection engineering and formal assurance. Finds defects in deployed software — DNS resolver logic, protocol interoperability, filesystem isolation — and reports them with reproduction, root cause, and code-level fixes; builds adversary-emulation environments that measure whether detections hold under attack, and expresses the results as machine-verifiable invariants rather than assertions. Applies the same discipline to agentic systems: what a platform mechanically guarantees about an agent’s authority, not what the agent is instructed to do.

Core technical competencies

DETECTION ENGINEERING	Sysmon process-tree instrumentation, Wazuh alert correlation, MITRE ATT&CK mapping, adversary emulation (Caldera), detection-gap analysis, MTTD measurement, Suricata, Velociraptor, TheHive / MISP
DEFECT DISCOVERY	Source-level bug discovery (PHP, config-generation logic), protocol and header analysis (RFC 6750), chroot and filesystem isolation failures, packet capture and socket analysis, formal reports with reproduction and code-diff fixes
SYSTEMS & PLATFORM	Linux namespace / chroot isolation, Unbound DNS internals, Active Directory architecture, Xen / KVM, VLAN segmentation and port-mirror design, ARM64 / x86-64 binary triage (Ghidra)
FORMAL METHODS & ASSURANCE	Machine-checkable invariants, graph-reachability isolation proofs, capability-based authority models (seL4 / Microkit), statistical detection SLOs with Wilson intervals, Enforced / Declared / Open assurance ledgers
AGENTIC-SYSTEM SECURITY	Capability-floor analysis, App Intents authorization semantics, authorization-bounded agent design — capability packages, exactly-once tokens, point-of-use permit validation, network-disabled sandboxed execution
PROGRAMMING & TOOLING	Python (primary), C, Bash; pytest / ruff / mypy gates; PHP and Swift source analysis; Ed25519 signing and hash-chained evidence logs; local-inference (MLX) triage tooling

Selected findings

MAR 2026	OPNsense 26.1.5 — three interacting DNS-stack bugs REPORTED UPSTREAM Three independent defects in OPNsense’s Unbound integration, found by PHP source review and confirmed via packet capture, socket analysis, and config-generation tracing. Together they compound to total DNS failure. Three formal reports with reproduction, root cause, and code-diff fixes submitted 2026-03-27. Python module loaded unconditionally. <code>\$module_config</code> initialised with <code>python</code> regardless of DNSBL state, causing <code>SERVFAIL</code> in <code>chroot</code> with no active blocklist. Fix: guard on enabled blocklist entries. chroot hardcoded to <code>/var/unbound</code>. The restricted filesystem lacks device nodes for outbound UDP; confirmed by <code>tcpdump</code> (zero outbound packets) and repeated forwarder timeouts. Heredoc blank line. <code>forward-addr</code> directives emitted outside the <code>forward-zone</code> block and silently ignored — forwarders never configured despite correct GUI settings.
FEB 2026	TheHive 5 / MISP 2.4 — authorization-header incompatibility WORKAROUND VALIDATED Root-caused via packet capture and HTTP trace: TheHive sends <code>Authorization: Bearer <key></code> per RFC 6750; MISP 2.4 expects the raw key and returns HTTP 403 behind a misleading “check your API key” message. Validated an Apache <code>mod_headers</code> rewrite that strips the prefix; proposed a configurable <code>authScheme</code> parameter upstream. Report TH-MISP-001.

APR 2026

LAB EVIDENCE

OP-002 — live adversary emulation and telemetry-gap analysis

- Caldera 5.x emulation of SMB lateral movement across a three-host AD domain; surfaced five detection gaps, two critical, which drove structural architecture changes.
- **GAP-001, beacon-masquerade survival.** splunkd.exe executing inside the wazuh-agent.exe (SYSTEM) process tree generated no alert — a trusted parent gave unconditional child-execution cover.
 - **GAP-002, Sysmon EID 1 / EID 11 dissociation.** A PowerShell file-create was captured while its process-create was absent, making command-line arguments unrecoverable; root-caused to ProcessCreate rule ordering.
 - Measured 27-second MTTD across four correlated Wazuh alerts; promoted invariant INV-14 (a Domain-Admin action produces a correlated Windows Security event 4672 within 60 s) from Roadmap to Enforced.

JUL 2026

PLATFORM GUARANTEE

Apple Intelligence agentic surface — compiler-enforced authentication floor

Verified that the Swift compiler enforces a build-time minimum authentication policy for schema-conforming App Intents (.requiresLocalDeviceAuthentication for sensitive intents such as .photos.deleteAssets) — a defensive property, reported as a platform guarantee, not a vulnerability. Reverse-engineered the Capture Data diagnostic binary (Ghidra, arm64): all nine gather* routines dispatch externally, relocating the attack surface to the dispatch boundary.

§ 4 ARCHITECTURE

Frameworks and architecture

OSIM v2.0

PUBLISHED

Adversarially-validated detection architecture

18 machine-verifiable invariants, a four-level detection-confidence framework, and a compounding adversary-cost model over a six-stage kill chain. Identified that Tier-0 access and DNS exfiltration carry zero dwell time because no network path exists — structural elimination over detection speed. Six deception assets placed at exact privilege boundaries, each a zero-false-positive event.

AEROS

SPECIFIED

Formally-specified research-lab architecture

A NIST-SP-style architecture for a single-operator lab, with topological containment stated as a machine-checkable invariant — negative reachability proven by parsing firewall, VLAN, and mirror-session configuration into a graph. Defines an assurance ledger whose downgrade operator automatically demotes any Enforced claim lacking a resolvable check and a recent passing result.

AEROS-0

IN PROGRESS

Capability-confined execution on seL4

A workload executor on the seL4 microkernel in which a component’s authority is exactly its capability set. The static confinement checker (Python) is built and running: it parses a Microkit system description into an influence graph and decides mediation and authority invariants before boot, and found two specification defects on first execution. Runtime demonstrator in progress.

§ 5 IN DEVELOPMENT

In development

- Planned or in early execution. No findings are claimed and no implementation is complete.
- **Kernel vulnerability research environment** — reproducible Linux-kernel lab (QEMU/KVM, GDB, KASAN/UBSAN, Syzkaller).
 - **ARM64 memory-corruption research** — PAC, MTE, and SLUB allocator interaction.
 - **Capability-floor lab** — testbed for whether an agentic tool-use stack mechanically bounds the agent’s capability set. Specification complete; implementation not begun.
 - **Firmware reverse engineering** — embedded-image analysis (Ghidra, Binwalk), boot-chain trust, UEFI/OVMF Secure Boot assumptions.

§ 6 BACKGROUND

Experience and education

2023 – PRESENT

Independent security researcher — I-XI Labs

Full-time independent research: the findings, frameworks, and lab work above.

2015 – 2023

Restaurant operations — Houston, TX

Facilitated memories with as much intention as the chef placed into the dishes he created. Worked continuously across both degree programs, transitioning to full-time security research in 2023.

SAN JACINTO COLLEGE

A.S. Computer Science (2021–2023) · A.A. Business Administration (2015–2020)